PHY-112 | Principles of Physics-2

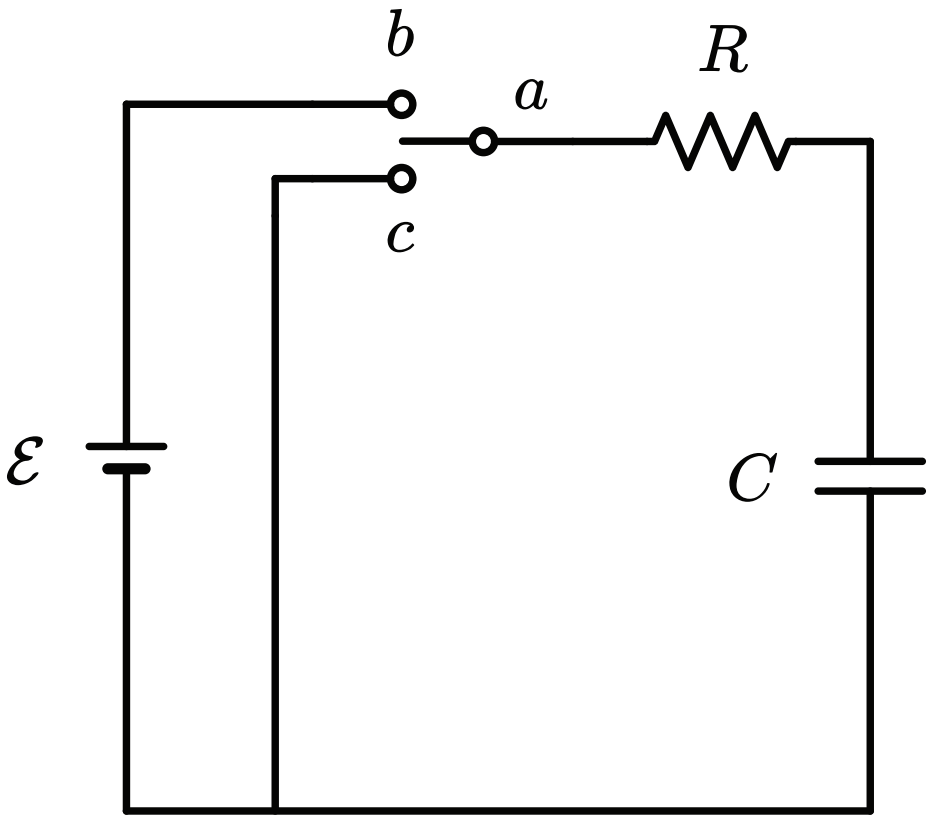
Akiful Islam (AZW)
Spring 2025 | Class #14

Department of Mathematics & Natural Sciences

BRAC UNIVERSITY
Inspiring Excellence

Series RC Circuit

R and *C* connected in series

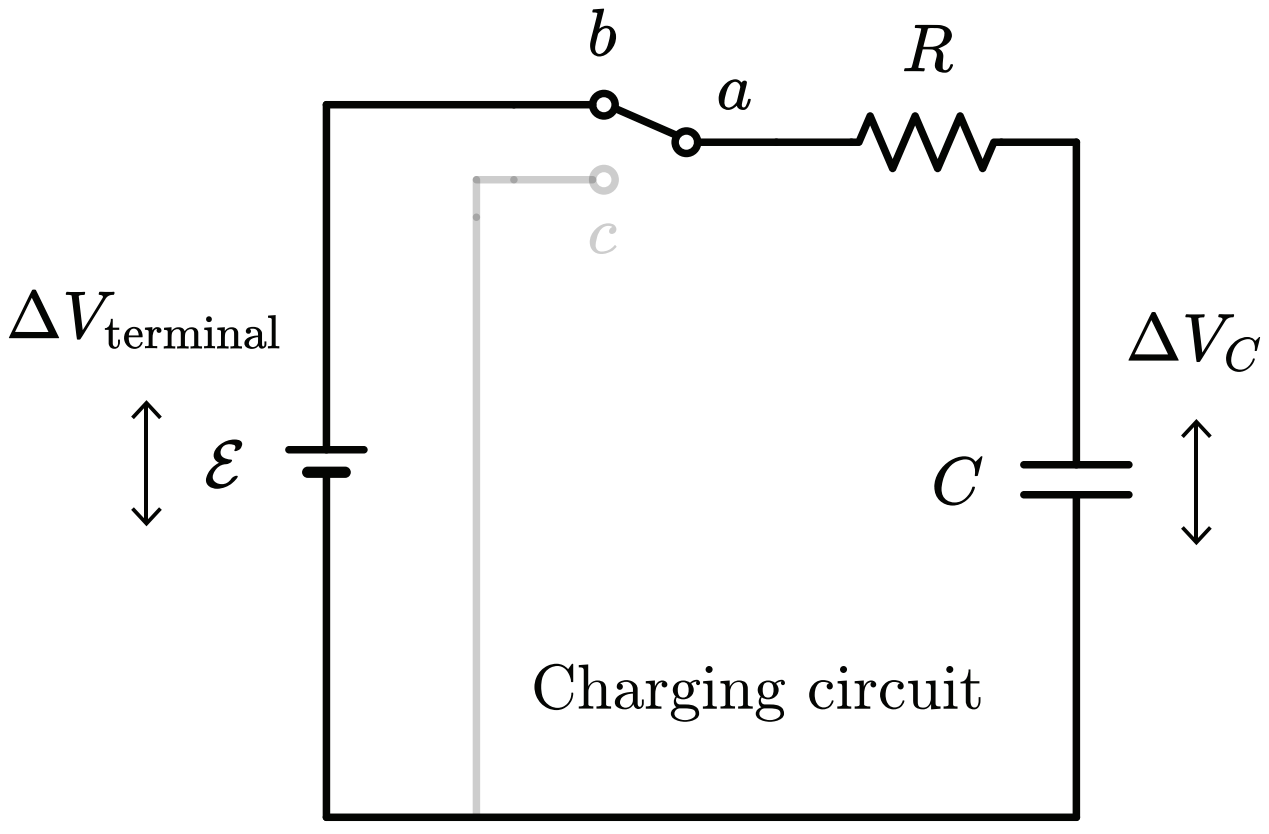


Charging a Capacitor

BRAC UNIVERSITY
Inspiring Excellence

Charging Circuit

Connect *a* with *b* node



Charging Capacitor: Growth of Charge

The rise is exponential in behavior

- Excess charge stored at the capacitor plates:

$$q(t) = C\mathcal{E}\left(1 - e^{-\frac{t}{RC}}\right).$$

  where $\tau_C = RC$ is the capacitive time constant and has the dimensions of time.
- Before charging: At $t = 0$:

$$q(t) = C\mathcal{E}\left(1 - e^{-\frac{0}{RC}}\right) = 0.$$

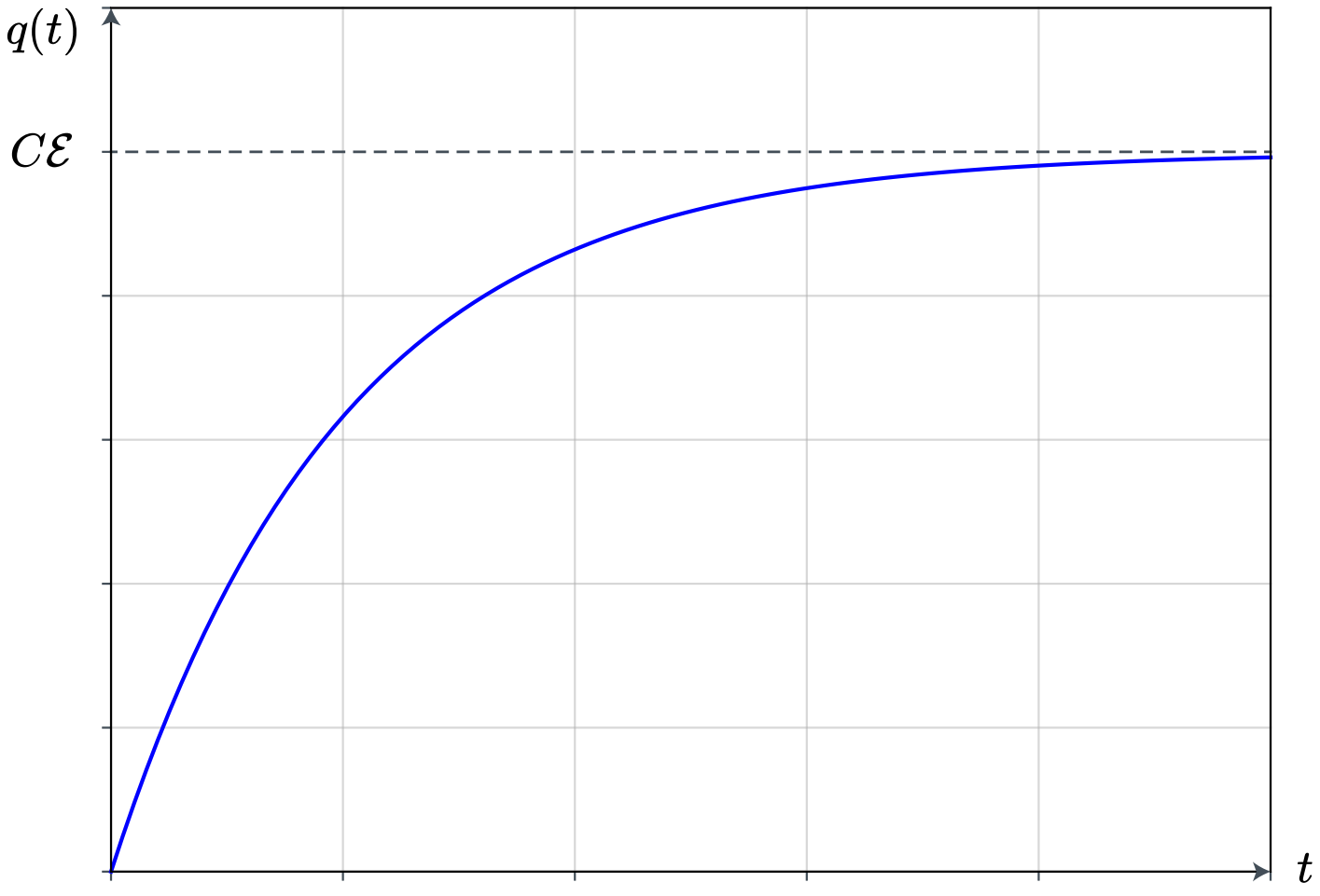
- After charging: At $t \to \infty$:

$$q(t) = C\mathcal{E}\left(1 - e^{-\frac{\infty}{RC}}\right) = C\mathcal{E}.$$

Charging Capacitor: Growth of Charge

The rise is exponential in behavior

Charging Phase



Charging Capacitor: Decay of Current

The drop is exponential in behavior

The transient current/charging current during the charging period of the capacitor.

$$i(t) = \frac{dq}{dt} = \frac{\mathcal{E}}{R} e^{-\frac{t}{RC}}.$$

At $t = 0$:

$$i(t) = \frac{\mathcal{E}}{R} e^{-\frac{0}{RC}} = \frac{\mathcal{E}}{R}.$$

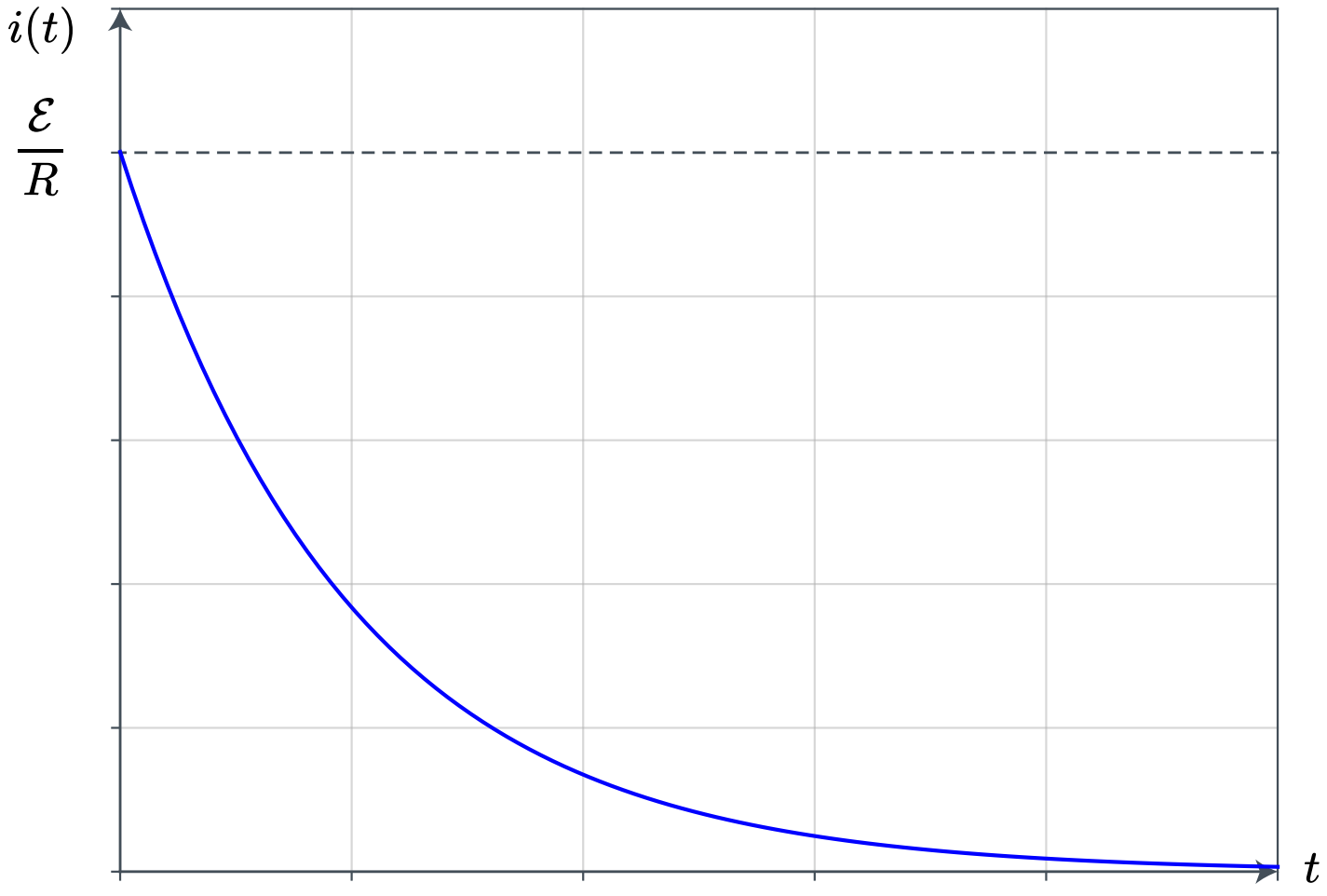
At $t \to \infty$:

$$i(t) = \frac{\mathcal{E}}{R} e^{-\frac{\infty}{RC}} = 0.$$

Charging Capacitor: Decay of Current

The drop is exponential in behavior

Charging Phase



Charging Capacitor: Decay of Resistive Voltages

The drop is exponential in behavior

The resistive voltage can be found by manipulating:

$$V_R^{\text{charging}}(t) = i(t)R = \mathcal{E}e^{-\frac{t}{RC}}.$$

At $t = 0$:

$$V_R^{\text{charging}}(t) = \mathcal{E}\left(1 - e^{-\frac{0}{RC}}\right) = \mathcal{E}.$$

At $t \to \infty$:

$$V_R^{\text{charging}}(t) = \mathcal{E}\left(1 - e^{-\frac{\infty}{RC}}\right) = 0.$$

Charging Capacitor: Growth of Capacitive Voltages

The growth is exponential in behavior

The capacitive voltage can be found by:

$$V_C^{\text{charging}}(t) = \frac{q(t)}{C} = \mathcal{E}\left(1 - e^{-\frac{t}{RC}}\right).$$

At $t = 0$:

$$V_C^{\text{charging}}(t) = \mathcal{E}\left(1 - e^{-\frac{0}{RC}}\right) = 0.$$

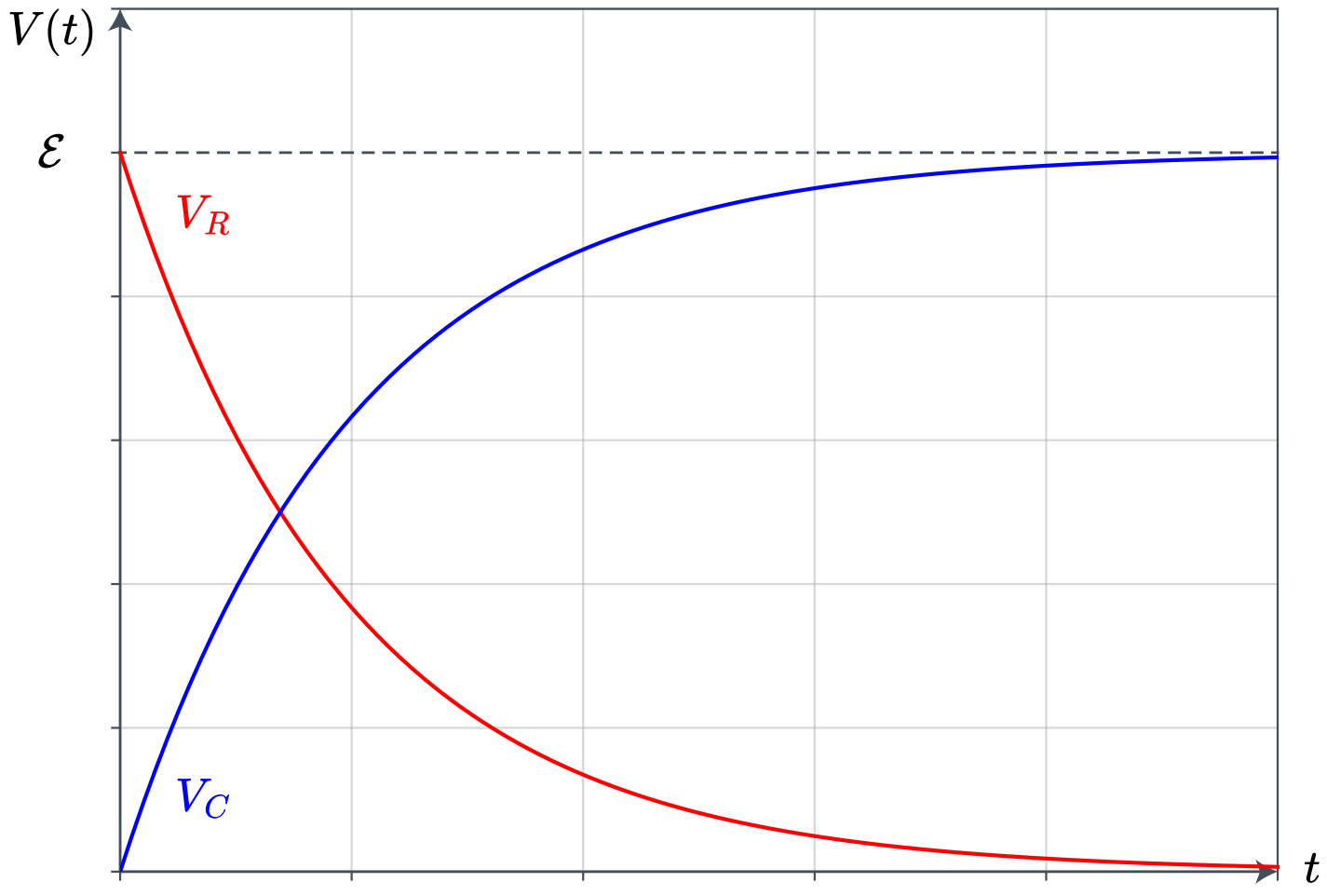
At $t \to \infty$:

$$V_C^{\text{charging}}(t) = \mathcal{E}\left(1 - e^{-\frac{\infty}{RC}}\right) = \mathcal{E}.$$

Charging Capacitor: Growth/Decay of Voltages

The rise/drop is exponential in behavior

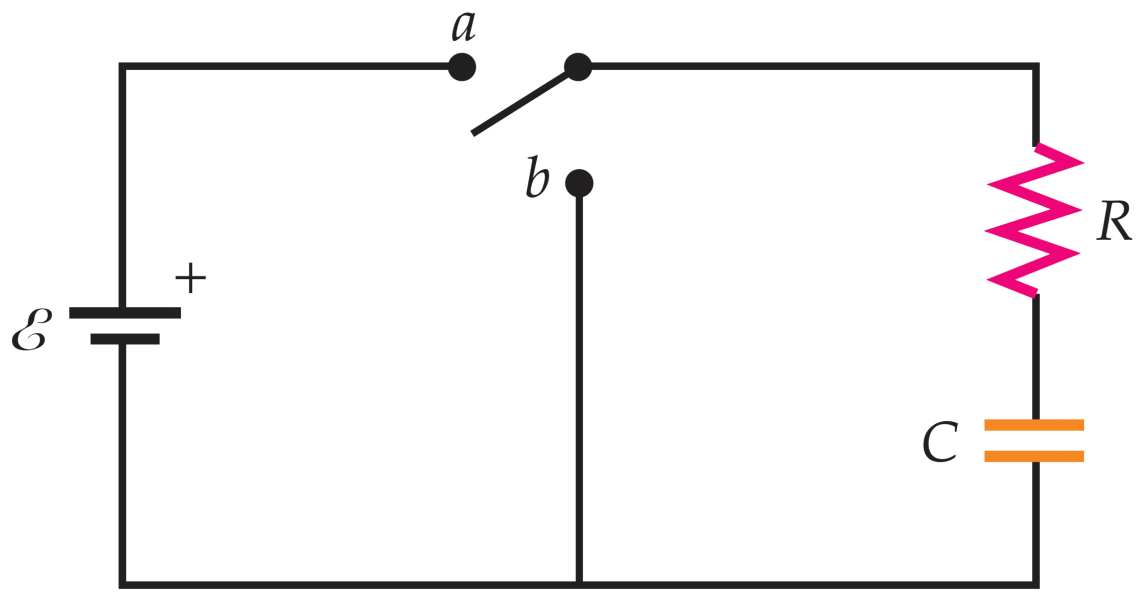
Charging Phase



Incepting Ideas (1)

Hint: Follow the charging equaiton list

Q: For the circuit shown, $C = 6\ \mu F$, $\mathcal{E} = 100\ V$ and $R = 500\ \Omega$. After having been at contact a for a long time, the switch throw is rotated to contact b. (a) Find the capacitive time constant of the circuit. (b) Find the charge at the capacitor plates and current in the circuit at $t = 1.5\tau_C$. (c) Measure the time during charging, when the capacitive and resistive voltage matches each other.

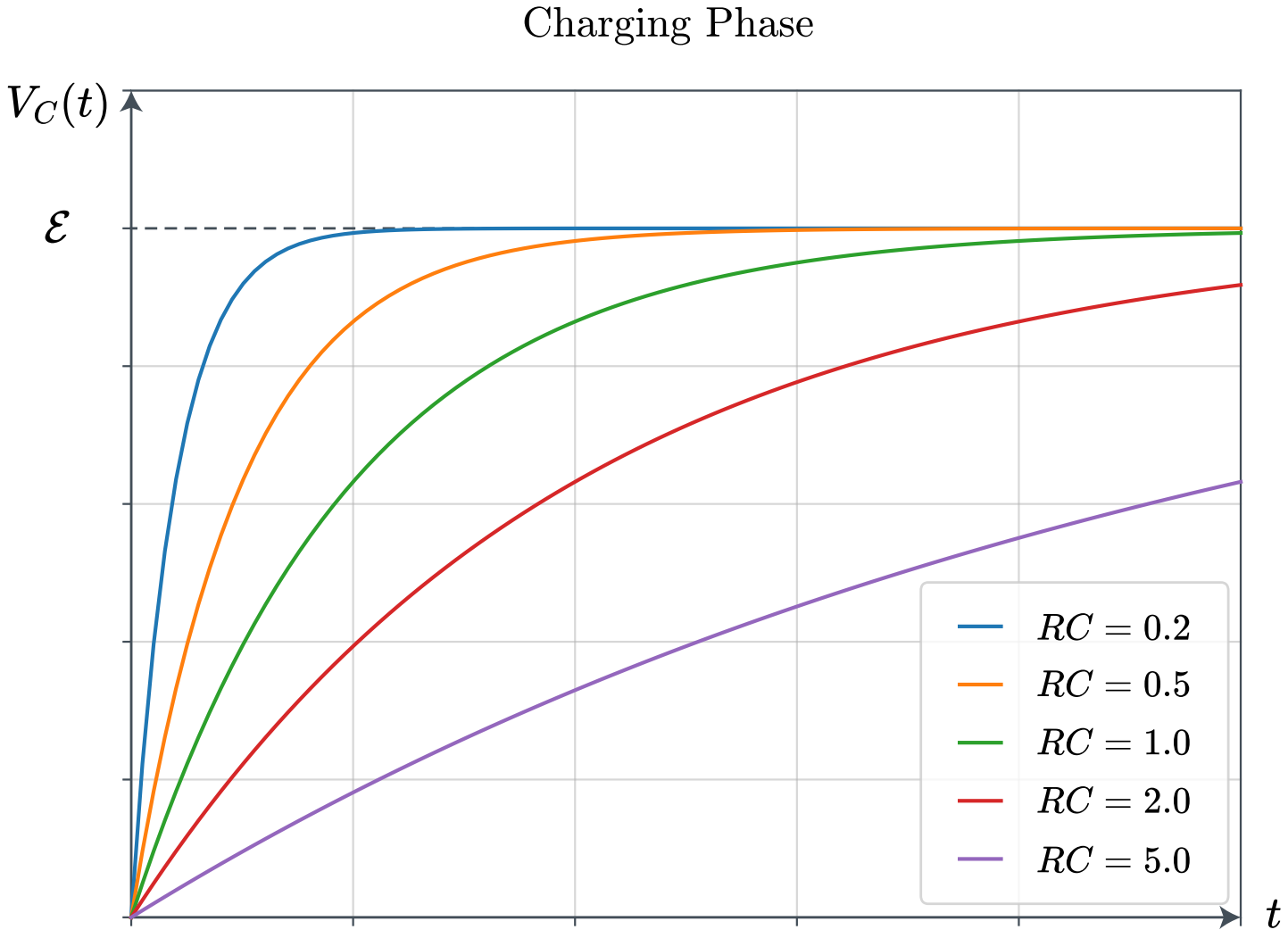


What's up with this Time Constant?

BRAC UNIVERSITY
Inspiring Excellence

Effects of τ_C in Charging/Discharging

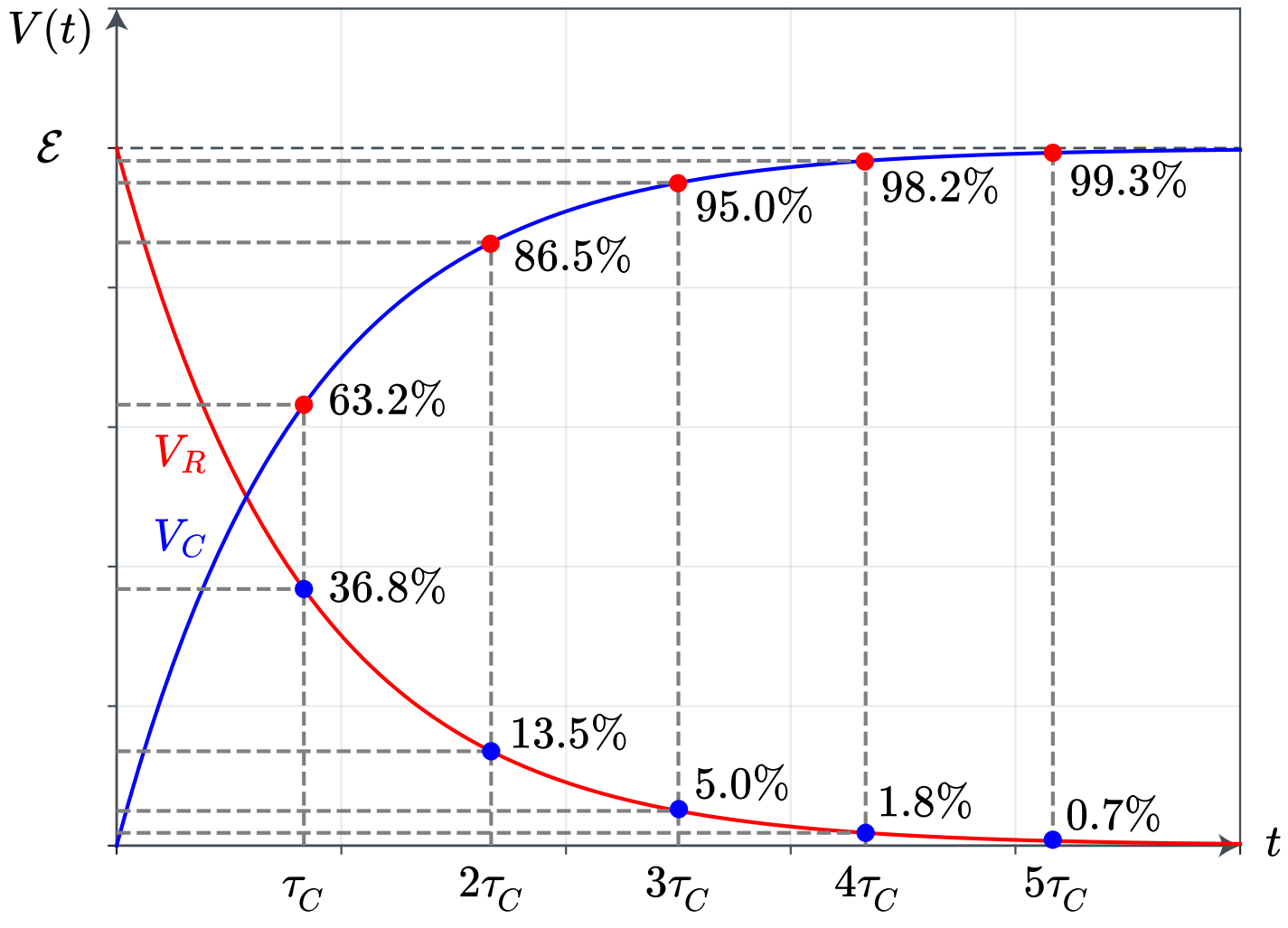
Charge/Discharging rate is inverse to the amount of τ_C



How charging/discharging looks

In terms of τ_C

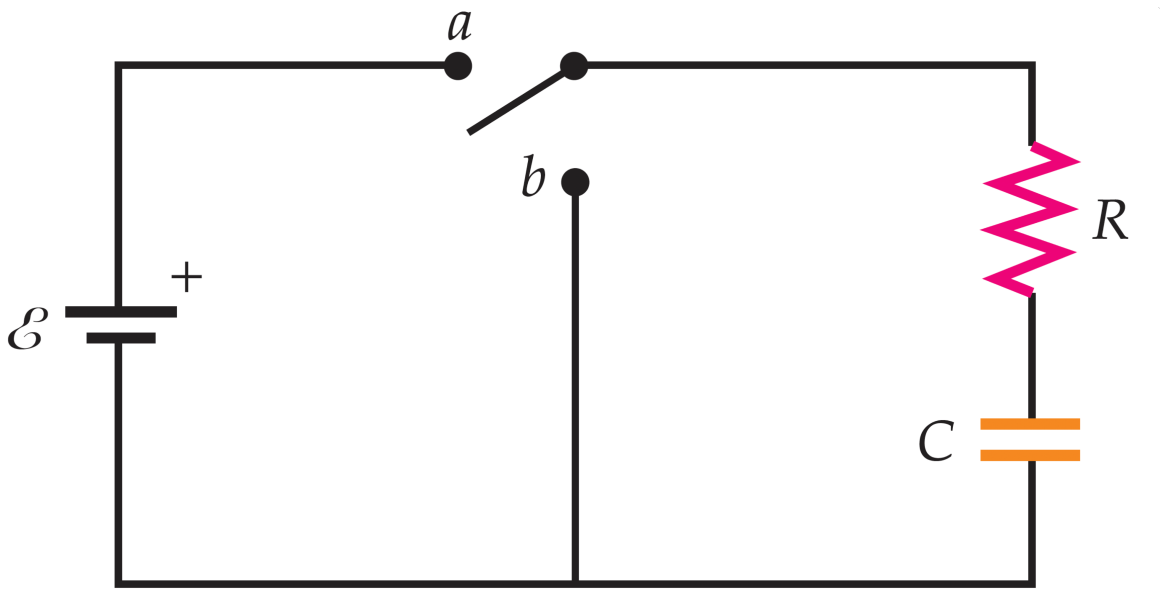
Charging Phase



Incepting Ideas (2)

Hint: Follow the charging equaiton list. Again!

Q: For the circuit shown, $C = 6\ \mu F$, $\mathcal{E} = 100\ V$ and $R = 500\ \Omega$. After having been at contact a for a long time, the switch throw is rotated to contact b. Find the values of voltages at times that are multiple of τ_C. Go until $5\tau_C$.

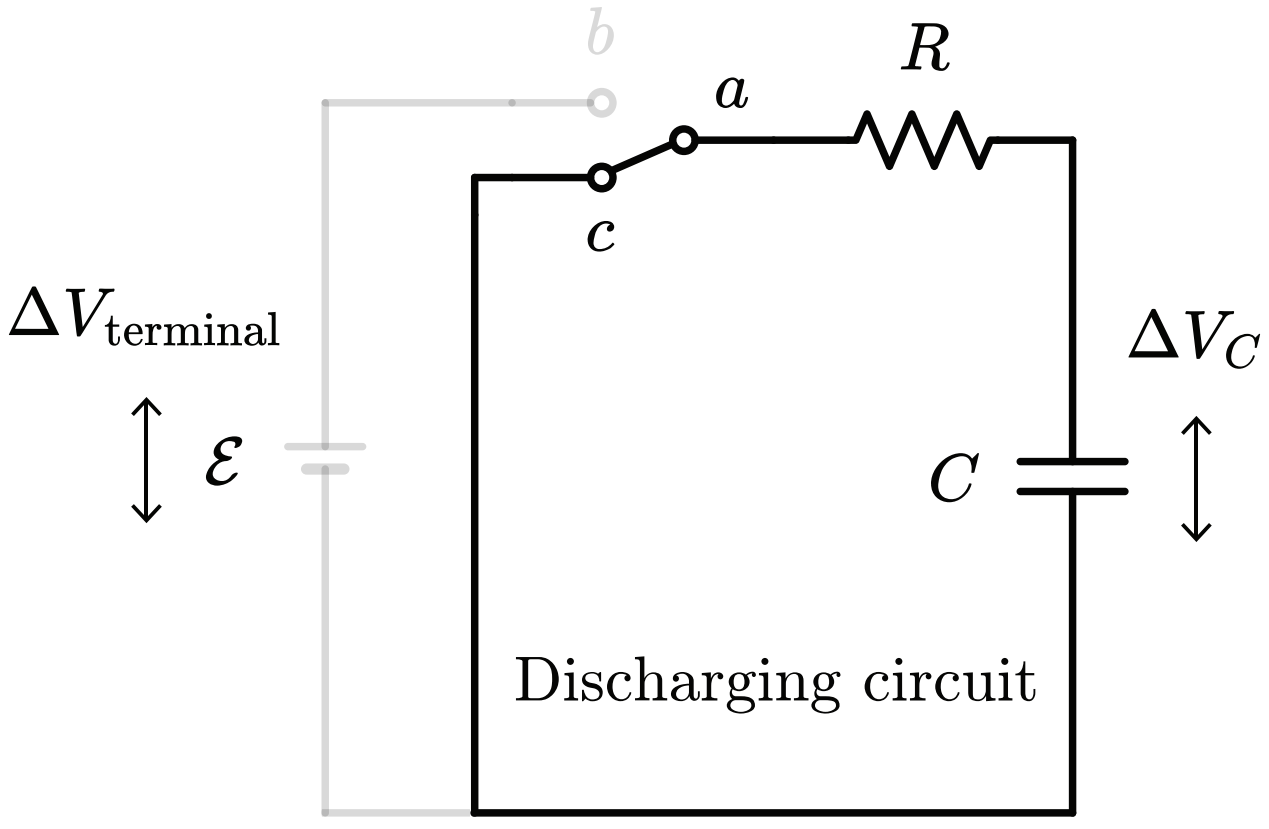


Discharging a Capacitor

BRAC UNIVERSITY
Inspiring Excellence

Disharging Circuit

Connect *a* with *c* node



Discharging Capacitor: Decay of Charge

The drop is exponential in behavior

- Excess charge discharges from the capacitor plates:

$$q(t) = Q_0 e^{-\frac{t}{RC}}.$$

- Before charging: At $t = 0$:

$$q(t) = Q_0 e^{-\frac{t}{RC}} = Q_0.$$

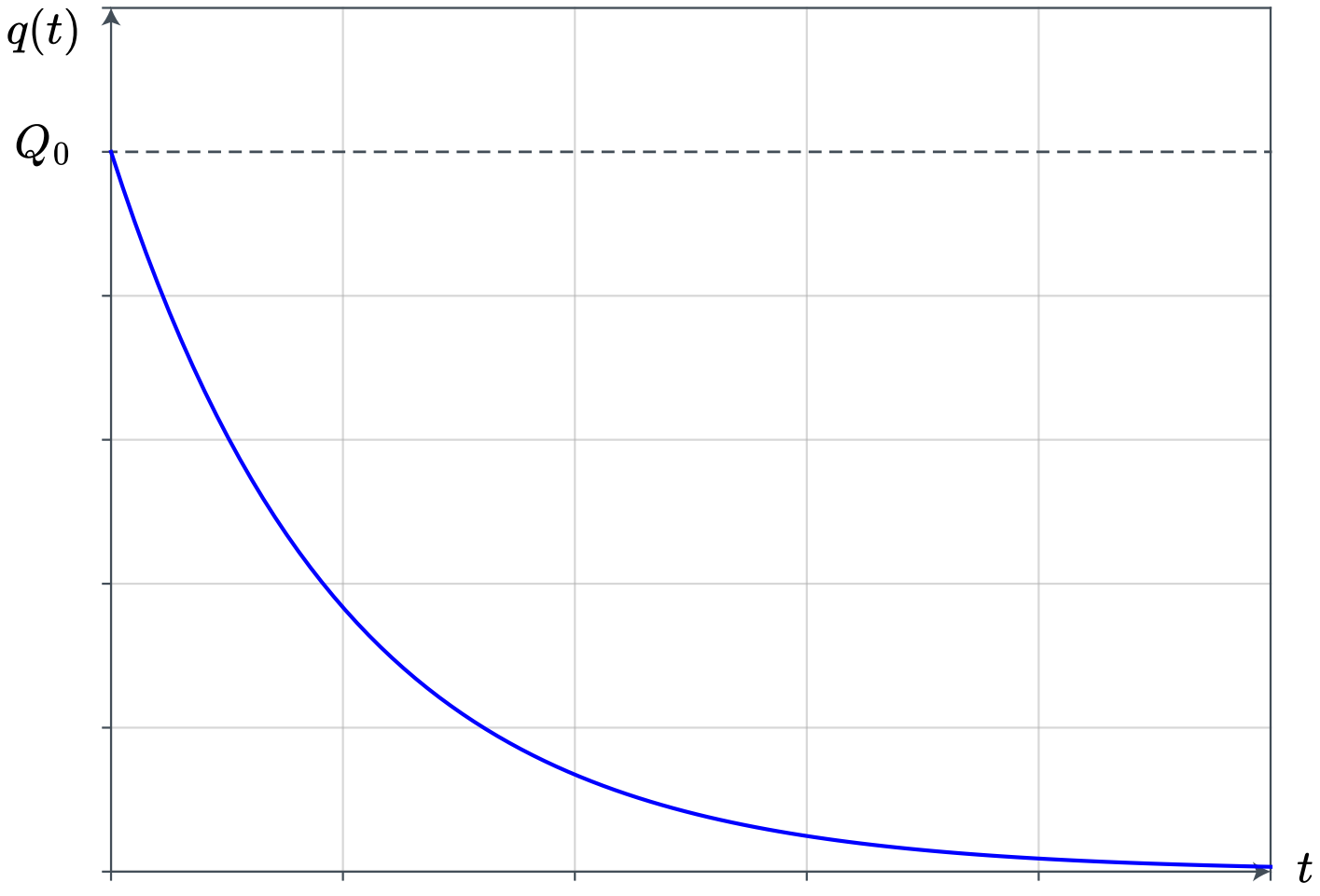
- After charging: At $t \to \infty$:

$$q(t) = Q_0 e^{-\frac{\infty}{RC}} = 0.$$

Discharging Capacitor: Decay of Charge

The drop is exponential in behavior

Discharging Phase



Discharging Capacitor: Decay of Current

The drop is exponential in behavior

The transient current/discharging current during the discharging period of the capacitor.

$$i(t) = \frac{dq}{dt} = -\frac{Q_0}{RC} e^{-\frac{t}{RC}}.$$

At $t = 0$:

$$i(t) = -\frac{Q_0}{RC} e^{-\frac{0}{RC}} = -\frac{Q_0}{RC}.$$

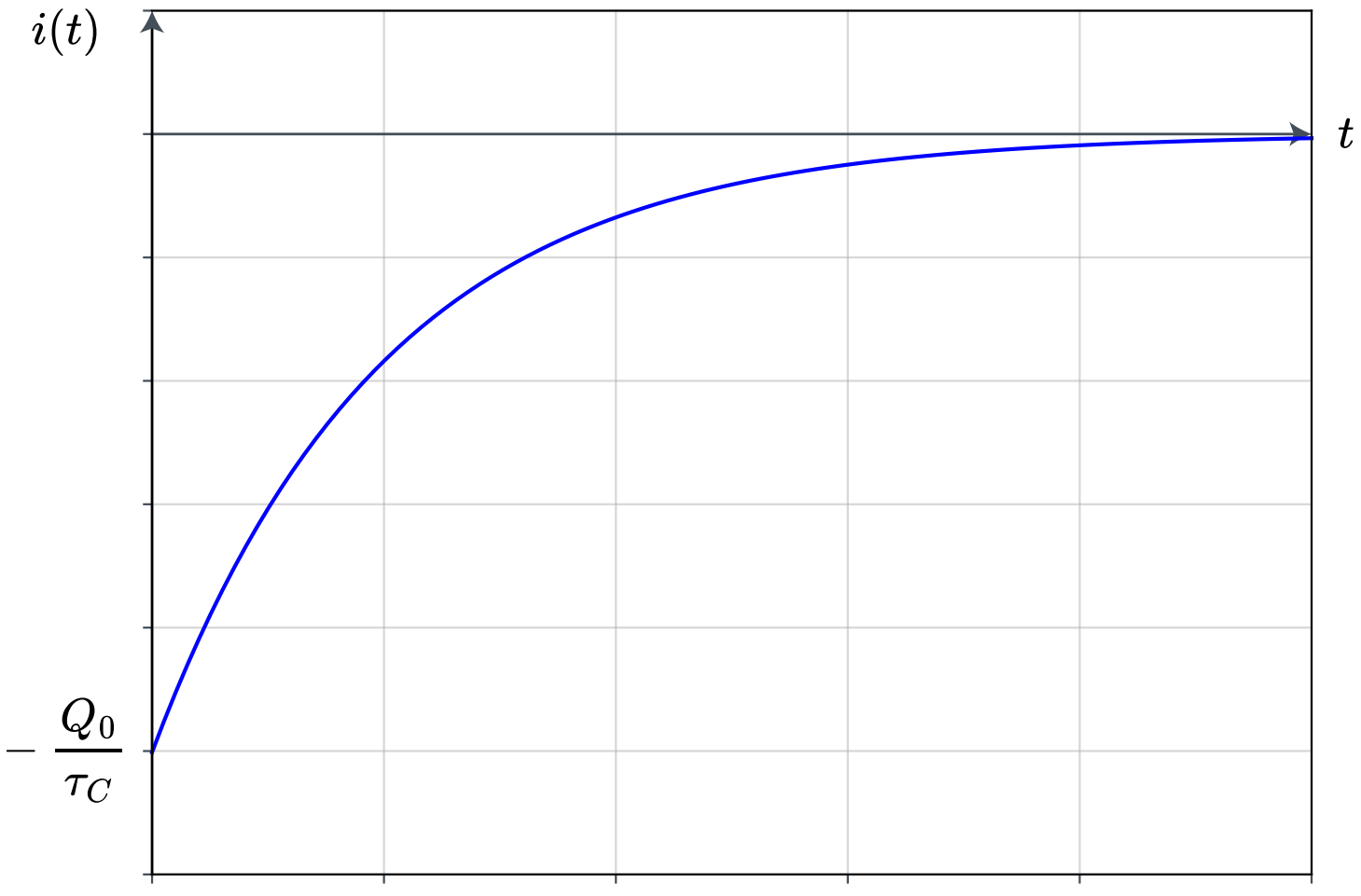
At $t \to \infty$:

$$i(t) = -\frac{Q_0}{RC} e^{-\frac{\infty}{RC}} = 0.$$

Discharging Capacitor: Decay of Current

The drop is exponential in behavior

Discharging Phase



Discharging Capacitor: Decay of Resistive Voltage

The drop is exponential in behavior

The resistive voltage can be found by:

$$V_R^{\text{discharging}}(t) = -\frac{q(t)}{C} = \frac{Q_0}{C} e^{-\frac{t}{RC}}.$$

At $t = 0$:

$$V_R^{\text{discharging}}(t) = -\frac{Q_0}{C} e^{-\frac{0}{RC}} = -\frac{Q_0}{C}.$$

At $t \to \infty$:

$$V_R^{\text{discharging}}(t) = -\frac{Q_0}{C} e^{-\frac{\infty}{RC}} = 0.$$

Discharging Capacitor: Decay of Capacitive Voltage

The drop is exponential in behavior

The capacitive voltage can be found by:

$$V_C^{\text{discharging}}(t) = \frac{q(t)}{C} = \frac{Q_0}{C} e^{-\frac{t}{RC}}$$

At $t = 0$:

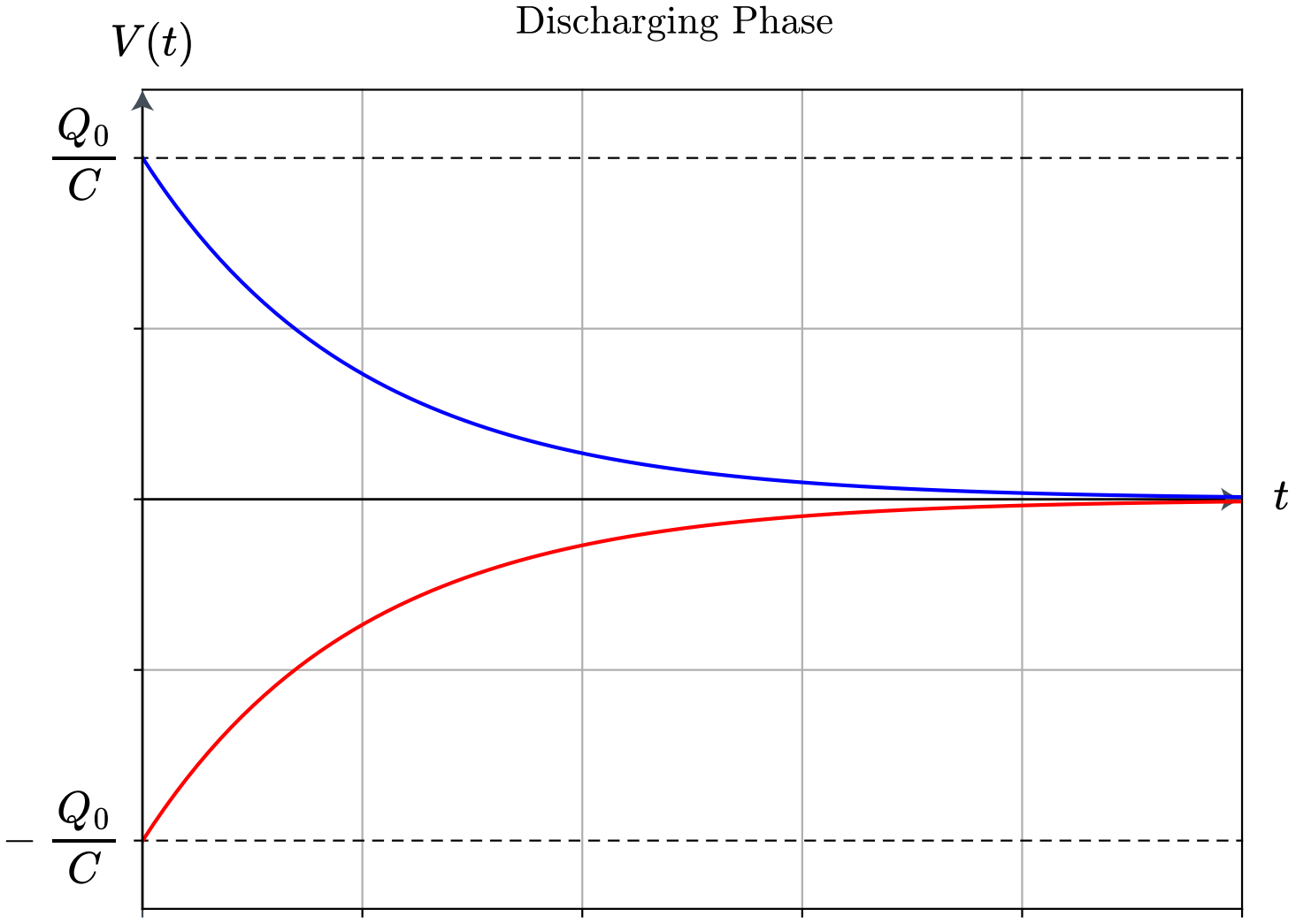
$$V_C^{\text{discharging}}(t) = \frac{Q_0}{C} e^{-\frac{0}{RC}} = \frac{Q_0}{C}.$$

At $t \to \infty$:

$$V_C^{\text{discharging}}(t) = \frac{Q_0}{C} e^{-\frac{\infty}{RC}} = 0.$$

Discharging Capacitor: Decay of Voltages

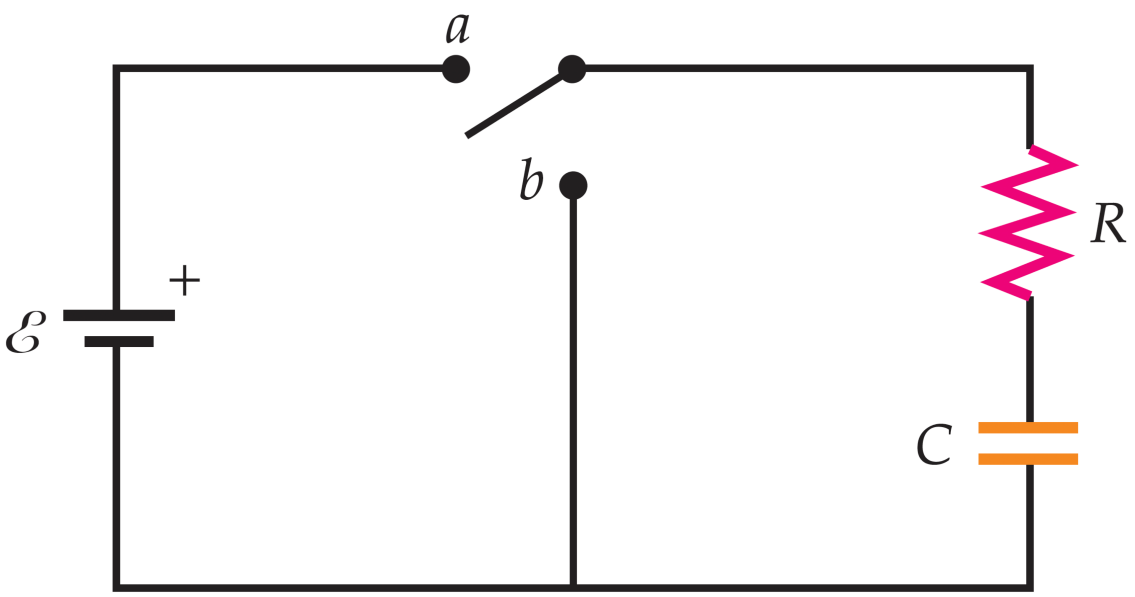
Their drop is exponential in behavior



Incepting Ideas (3)

Hint: Follow the charging equaiton list

Q: For the circuit shown, $C = 6\ \mu F$, $\mathcal{E} = 100\ V$ and $R = 500\ \Omega$. After having been at contact a for a long time, the switch throw is rotated to contact b. (a) What is the current just after the switch throw is rotated to contact b? (b) How much charge is on the upper plate of the capacitor 6 ms after the switch throw is rotated to contact.



Energy stored during Charging & Discharging

BRAC UNIVERSITY
Inspiring Excellence

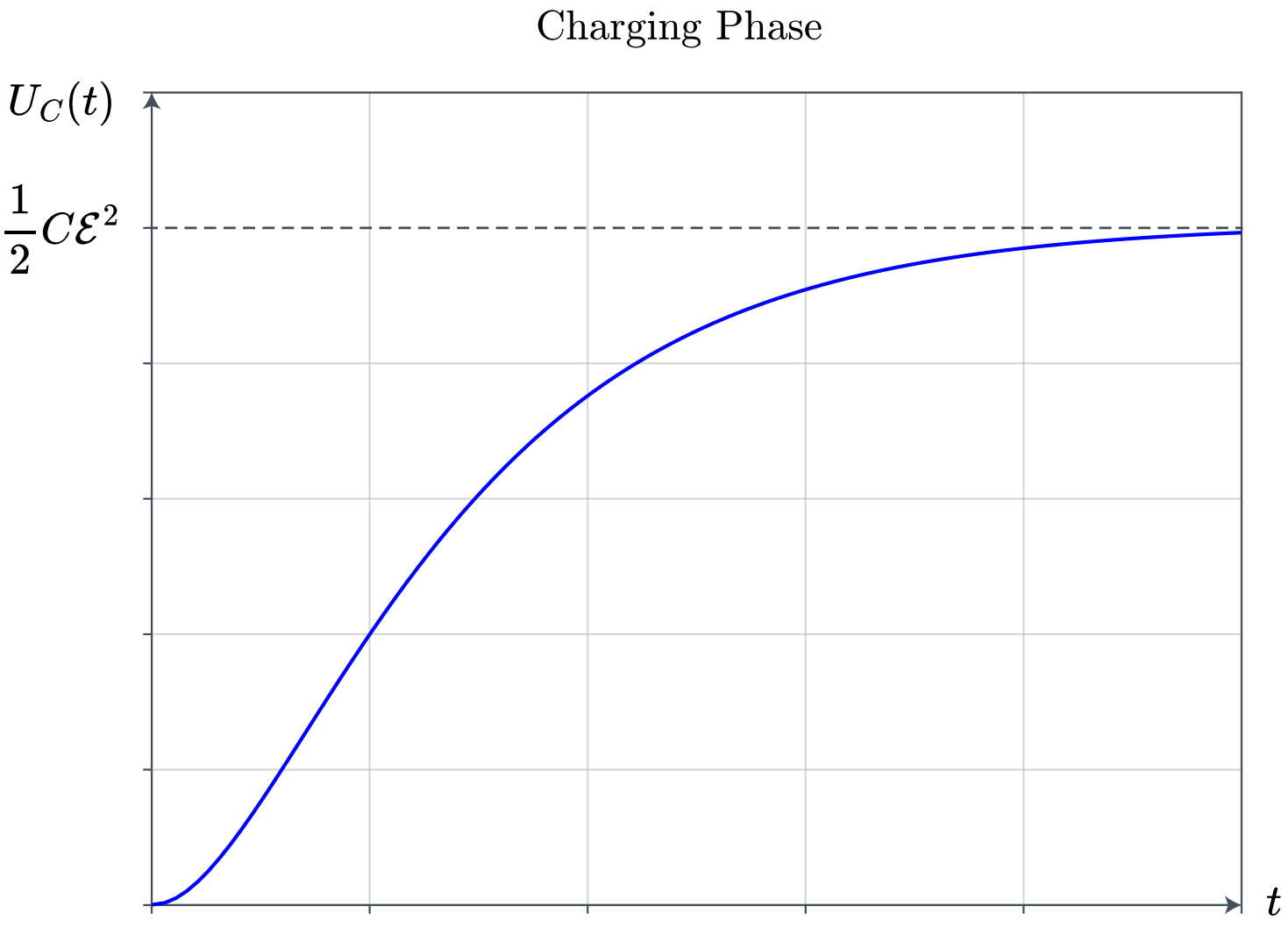
Charging Capacitor: Growth of Energy

This growth is exponential as well

$$
\begin{aligned}
U_C^{\text{charging}} &= \frac{1}{2} C \left[V_C^{\text{charging}}(t) \right]^2 \\
&= \frac{1}{2} C \left[\mathcal{E} \left(1 - e^{-\frac{t}{RC}} \right) \right]^2 \\
&= \frac{1}{2} C \mathcal{E}^2 \left(1 - e^{-\frac{t}{RC}} \right)^2 .
\end{aligned}
$$

Charging Capacitor: Growth of Energy

This growth is exponential as well



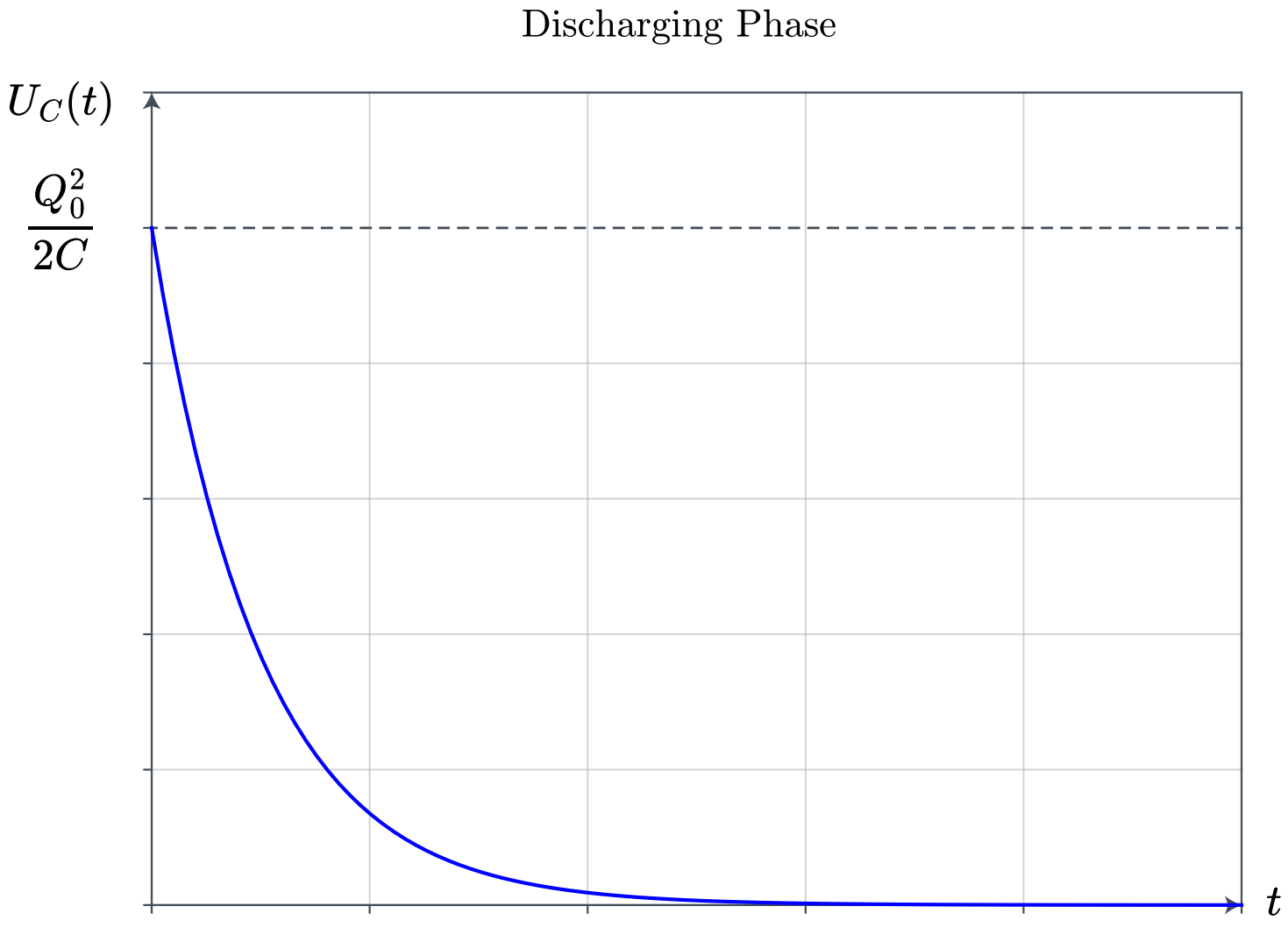
Discharging Capacitor: Decay of Energy

This decay is exponential as well

$$
\begin{aligned}
U_C^{\text{discharging}} &= \frac{1}{2} C \left[V_C^{\text{discharging}}(t) \right]^2 \\
&= \frac{1}{2} C \left[\frac{Q_0}{C} e^{-\frac{t}{RC}} \right]^2 \\
&= \frac{Q_0^2}{2C} e^{-\frac{2t}{RC}}.
\end{aligned}
$$

Discharging Capacitor: Decay of Energy

This decay is exponential as well

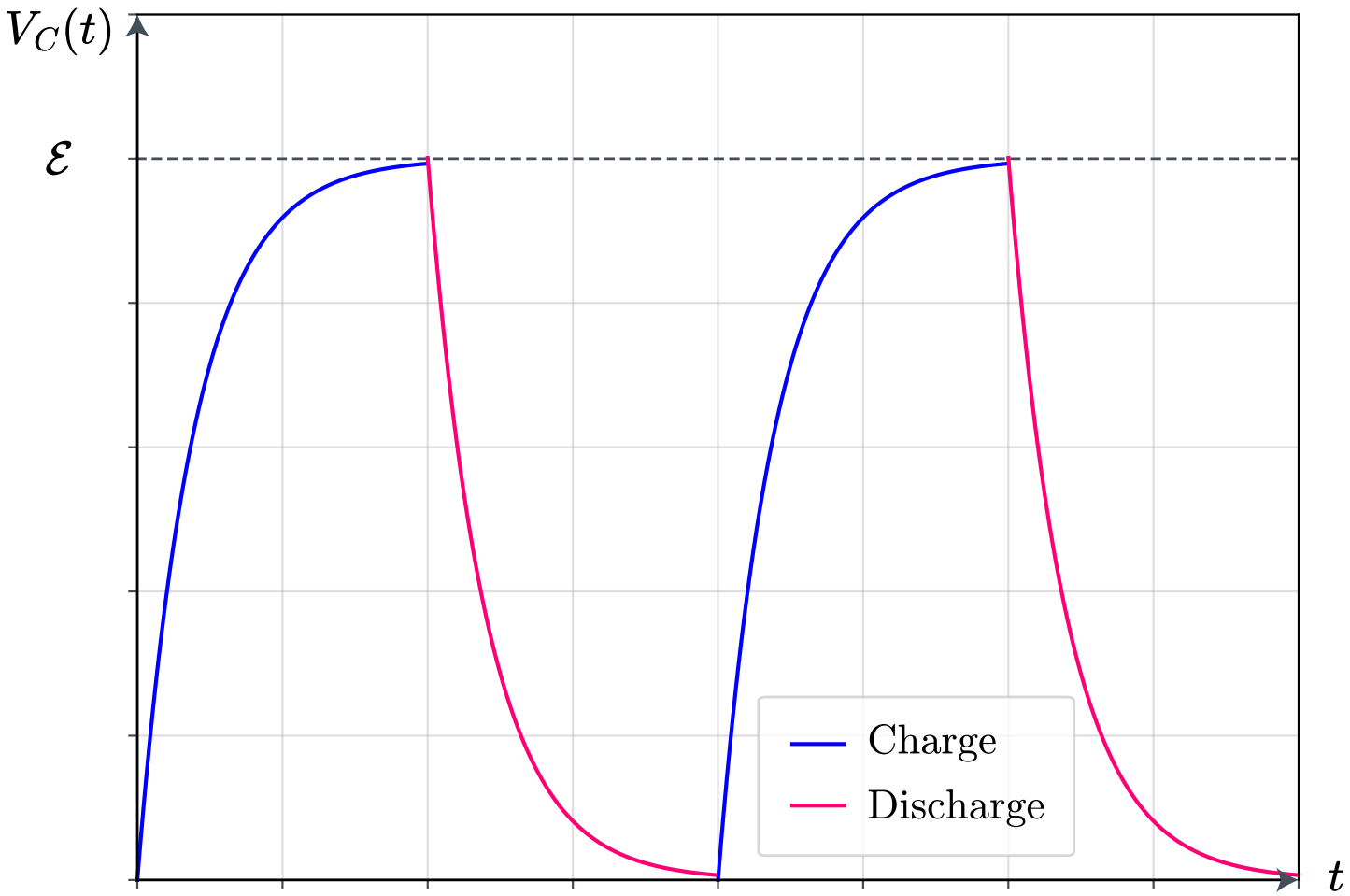


Experimental Work You did

BRAC UNIVERSITY
Inspiring Excellence

Charging & Discharging Capacitor

You discharge with what you charged it to



Incepting Ideas (4)

Hint: Practice! Practice! Practice!

Some Problems to Practice at Home on this topic

Example Problem 26.12, 26.13 p-890 | Young-Freedman
Exercise Problem 26.40, 26.42, 26.43, 26.46, p-901 | YF
Exercise Problem 57, 58, 59, 60, p-799 | RH

That is it for today!

BRAC
Inspiring Excellence